



Department of Computer Engineering

CSE5041 Database Design & Development
Project Report

Home Sharing Platform

Students:

ID	Name & Surname
2300003388	ZAID IYAD J. J. DWEEKAT
2400001392	MUSAEED MOHAMMED SALEH ALHAROGI

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SAMPLE

1 INTRODUCTION

1.1 PROJECT DESCRIPTION

The **Home Sharing Platform (HSP)** is a database system designed to store and manage all information related to users, property owners, property listings, bookings, payments, messaging, notifications, availability, amenities, photos, and reviews. The following data requirements were gathered during the requirements collection and analysis phase and must be represented in the enterprise database:

The platform allows individuals to register as users. Each user has a unique `User_ID`, and the system stores their first name, last name, email (unique), password, phone number, residential address, and a role (`Role_ID`) that defines their access level on the platform.

Some users can act as property owners. Each property listing is linked to exactly one owner (`Owner_ID`), and one owner can register one or more properties on the platform.

A property represents a home, apartment, or room available for booking. Each property has a unique `Property_ID` and stores important listing information such as title, description, maximum capacity, and daily price. Each property is linked to one location record (`Location_ID`) that stores detailed address information (country, city, street, building number, apartment number, and postal code). The platform also supports media and availability management by storing multiple property photos and marking unavailable dates for each property, including optional reasons (e.g., maintenance).

Users can browse available properties and make bookings. A booking represents a reservation made by a user for a specific property. Each booking has a unique `Booking_ID`, check-in date, check-out date, status, and calculated total price. A user can make multiple bookings, and each property can be booked many times, but the system must ensure that bookings for the same property do not overlap in dates.

Each booking requires a payment record. A payment has a unique `Payment_ID` and stores the payment amount, method, and payment date. Each booking is associated with exactly one payment to support tracking and verification of transactions.

After completing a stay, a user may leave a review for the property. Each review has a unique `Review_ID` and stores the rating (1–5), written feedback, and review date. A property may receive multiple reviews, and each review is linked to both the user who wrote it and the property it is about. The platform also supports owner interaction by allowing an optional reply to each review, where the reply includes text and a reply date.

To improve search, filtering, and property details, the platform manages amenities. Each amenity has a unique `Amenity_ID` with a name and type. Because a property can provide many amenities and the same amenity can exist in many properties, the system stores this relationship using an associative table that also records the quantity of each amenity provided per property.

In addition, the platform supports basic user communication and system interaction features. Users can send messages to each other through a messaging system, where each message stores the sender, receiver, message content, and timestamp. The system also generates notifications for users (e.g., booking updates or reminders), where each notification stores a message, status (such as unread/read), and the notification date.

Overall, the Home Sharing Platform database ensures efficient storage, retrieval, and management of all data required for property browsing, booking stays, managing listings, processing payments, handling availability, maintaining user-generated reviews and replies, supporting amenities and photos, and enabling messaging and notifications

2 ENTITY RELATIONAL MODEL

2.1 ENHANCED ER DIAGRAM

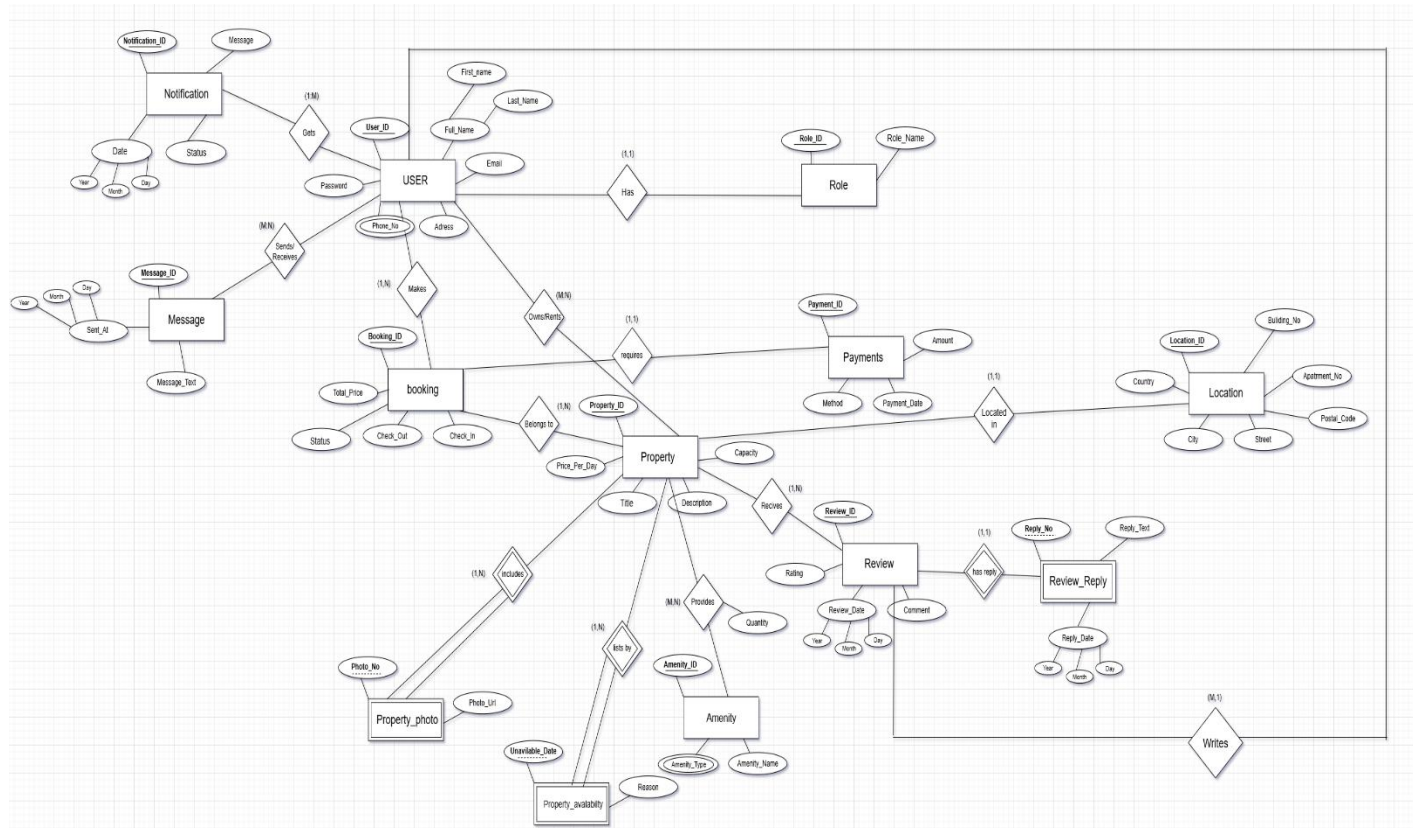


Figure 1: EER diagram of the Home Sharing Platform Database

2.2 RELATIONAL SCHEMA & MAPPING

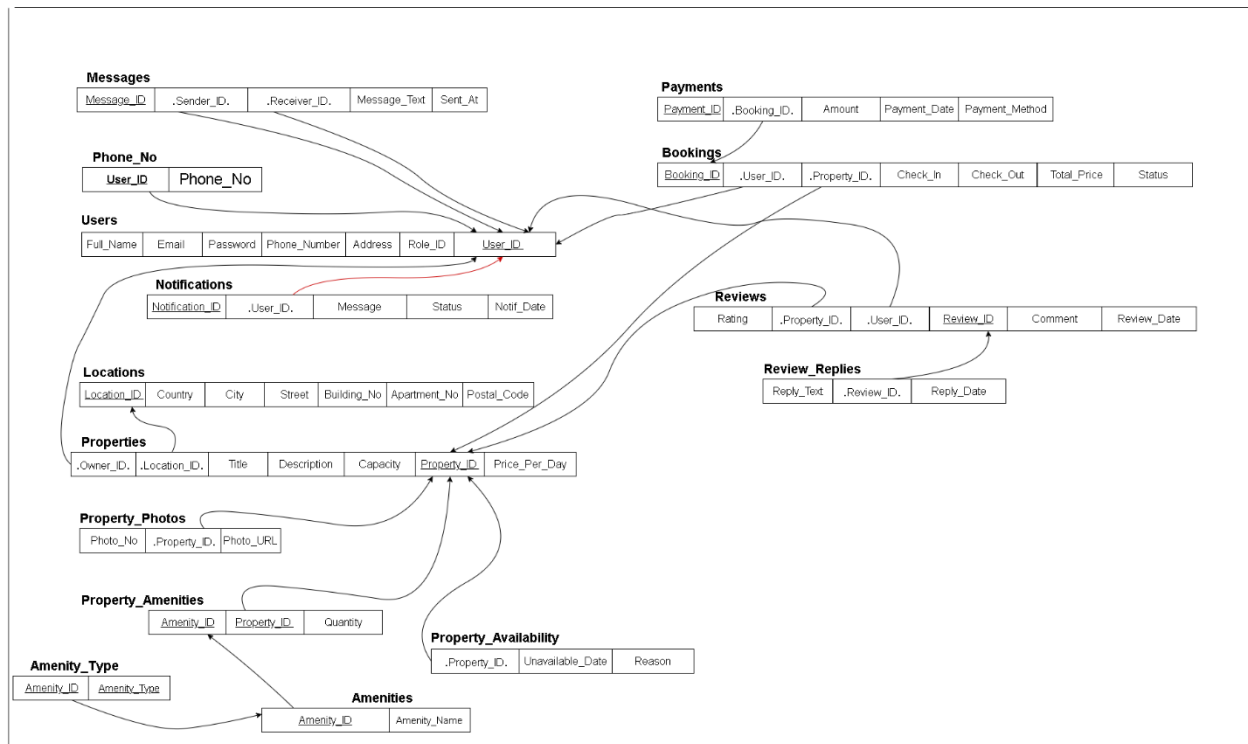


Figure 2: Relational schema of the **Home Sharing Platform** Database with arrows indicating referential integrity

3 NORMALIZATION

3.1 FUNCTIONAL DEPENDENCIES

FD1: Role_ID $\rightarrow$ Role_Name

FD2: User_ID $\rightarrow$ First_Name, Last_Name, Email, Password, Phone_No, Address, Role_ID

FD3: Email $\rightarrow$ User_ID, First_Name, Last_Name, Password, Phone_No, Address, Role_ID

FD4: Location_ID $\rightarrow$ Country, City, Street, Building_No, Apartment_No, Postal_Code

FD5: Amenity_ID $\rightarrow$ Amenity_Name, Amenity_Type

FD6: Property_ID $\rightarrow$ Owner_ID, Location_ID, Title, Description, Price_Per_Day, Capacity

FD7: Location_ID $\rightarrow$ Property_ID, Owner_ID, Title, Description, Price_Per_Day, Capacity

FD8: Booking_ID $\rightarrow$ User_ID, Property_ID, Check_In, Check_Out, Total_Price, Status

FD9: Payment_ID $\rightarrow$ Booking_ID, Amount, Method, Payment_Date

FD10: Booking_ID $\rightarrow$ Payment_ID, Amount, Method, Payment_Date

FD11: Review_ID $\rightarrow$ Property_ID, User_ID, Rating, Comment, Review_Date

FD12: Review_ID $\rightarrow$ Reply_Text, Reply_Date

FD13: Property_ID, Photo_No $\rightarrow$ Photo_Url

FD14: Property_ID, Unavailable_Date $\rightarrow$ Reason

FD15: Property_ID, Amenity_ID $\rightarrow$ Quantity

FD16: Message_ID $\rightarrow$ Sender_ID, Receiver_ID, Message_Text, Sent_At

FD17: Notification_ID $\rightarrow$ User_ID, Message, Status, Notif_Date

3.2 UN-NORMALIZED FORM

User_ID	First_Name	Last_Name	Email
Password	Phone_No	Address	Role_ID
Role_Name	Property_ID	Owner_ID	Location_ID
Title	Description	Price_Per_Day	Capacity
Location(Country, City, Street, Building_No, Apartment_No, Postal_Code)	Photo_No, Photo_Url	Unavailable_Date, Reason	Amenity_ID, Amenity_Name, Amenity_Type, Quantity
Booking_ID, Check_In, Check_Out, Total_Price, Status, Payment_ID, Amount, Method, Payment_Date	Review_ID, Rating, Comment, Review_Date, Reply_Text, Reply_Date	Notification_ID, Message, Status, Notif_Date}	Message_ID, Sender_ID, Receiver_ID, Message_Text, Sent_At

3.3 First Normal Form

User_ID	First_Name	Last_Name	Email	Password	Phone_No
Address	Role_ID	Role_Name			

Property_1NF

Property_ID	Owner_ID	Location_ID	Country	City	Street
Building_No	Apartment_No	Postal_Code	Title	Description	Price_Per_Day
Capacity					

Booking_1NF

Booking_ID	User_ID	Property_ID	Check_In	Check_Out	Total_Price
Status	Payment_ID	Amount	Method	Payment_Date	

Review_1NF

Review_ID	Property_ID	User_ID	Rating	Comment	Review_Date
Reply_Text	Reply_Date				

Property_Photo

Property_ID	Photo_No	Photo_Url
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Property_Availability

Property_ID	Unavailable_Date	Reason
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Property_Amenity_1NF

Property_ID	Amenity_ID	Amenity_Name	Amenity_Type	Quantity
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Messages

Message_ID	Sender_ID	Receiver_ID	Message_Text	Sent_At
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Notifications

Notification_ID	User_ID	Message	Status	Notif_Date
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3.4 Second Normal Form

User_ID	First Name	Last Name	Email	Password	Phone_No
Address	Role_ID	Role Name			

Property 2NF

Property_ID	Owner_ID	Location_ID	Country	City	Street
Building_No	Apartment_No	Postal_Code	Title	Description	Price_Per_Day
Capacity					

Booking 2NF

Booking_ID	User_ID	Property_ID	Check_In	Check_Out	Total_Price
Status	Payment_ID	Amount	Method	Payment_Date	

Review 2NF

Review_ID	Property_ID	User_ID	Rating	Comment	Review_Date
Reply_Text	Reply_Date				

Amenities

Amenity_ID	Amenity_Name	Amenity_Type
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Property Amenities

Property_ID	Amenity_ID	Quantity
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Property Photo

Property_ID	Photo_No	Photo_Url
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Property Availability

Property_ID	Unavailable_Date	Reason
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Messages

Message_ID	Sender_ID	Receiver_ID	Message_Text	Sent_At
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Notifications

Notification_ID	User_ID	Message	Status	Notif_Date
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3.5 Third Normal Form**Roles**

Role_ID	Role_Name
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Users

User_ID	First Name	Last Name	Email	Password	Phone_No
Address	Role_ID				

Locations

Location_ID	Country	City	Street	Building_No	Apartment_No
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Postal_Code					
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Amenities

Amenity_ID	Amenity_Name	Amenity_Type
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Properties

Property_ID	Owner_ID	Location_ID	Title	Description	Price_Per_Day
Capacity					

Bookings

Booking_ID	User_ID	Property_ID	Check_In	Check_Out	Total_Price
Status					

Payments

Payment_ID	Booking_ID	Amount	Method	Payment_Date
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Reviews

Review_ID	Property_ID	User_ID	Rating	Comment	Review_Date
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Review Reply

Review_ID	Reply_Text	Reply_Date
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Property Photo

Property_ID	Photo_No	Photo_Url
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Property Availability

Property_ID	Unavailable_Date	Reason
-------------	------------------	--------

Property Amenities

Property_ID	Amenity_ID	Quantity
-------------	------------	----------

Messages

Message_ID	Sender_ID	Receiver_ID	Message_Text	Sent_At
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Notifications

Notification_ID	User_ID	Message	Status	Notif_Date
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